same number, regardless of how much the outside universe has expanded in the meantime. If your original sphere contains a cluster of galaxies and some quantity of dark energy, in a billion years the amount of dark energy in that region will still be the same, so if it wasn't enough to mess up the galaxy cluster before, it won't be in the future. The balance between matter and dark energy in that sphere does not significantly change even as the rest of the cosmos seems to inexorably empty out.

This is reassuring. If you happen to be a clump of matter in the universe, and you would like to form a nice stable gravitationally bound galaxy, you can rest assured that once you get enough matter together to build something, dark energy won't ruin all that hard work.

Unless, that is, the dark energy is something more powerful than a cosmological constant.

As we discussed in the previous chapter, a cosmological constant is just one possibility for dark energy. All we really know about dark energy is that it's something that makes the universe expand faster. Or, more precisely, it has *negative pressure*. Negative pressure is a weird concept, because normally one thinks of pressure as something that pushes outward. But in Einstein's general-relativistic way of thinking about the universe, pressure is just another kind of energy, like mass, or radiation, and is thus gravitationally attractive. And in general relativity, gravitational attraction is just a consequence of the bending of space.

Remember the picture of the bowling ball creating a dent in the trampoline as an analogy for the effect of matter on the curvature of space? If you take general relativity into account, the dent is deeper if the ball is more massive, but also if it is hot, or if it has high internal pressure. So pressure, like other forms of energy, acts a lot like mass. From a gravitational perspective, pressure pulls. When you calculate the gravitational effect of a clump of gas, for instance, you have to factor in not just its mass, but its pressure, and both contribute to the gravitational impact that gas has on the stuff around it. In fact, the pressure contributes more to the spacetime curvature than the mass does.

What does that mean for something with *negative* pressure? If the pressure of some weird substance can be negative, it means that it can effectively *cancel out* the mass of the stuff, at least as regards its impact on the curving of spacetime. If you write down the pressure and density of